

Humanoid robot platform designed to walk faster and more safely

Engineered by a research team in the United States, a humanoid robot platform is designed to walk faster and more safely by combining advanced electronics, sensors, and a hybrid leg architecture. The system integrates multimodal

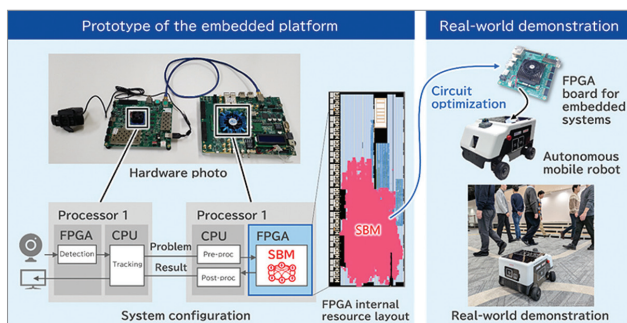


Yipu Chen (seated) and Zhaoyuan Gu (standing) on the CAREN treadmill system (Credit: Georgia Institute of Technology)

fall-detection electronics using inertial, proprioceptive, and acoustic sensors to detect instability and protect the robot during experiments. Its HybridLeg design blends serial and parallel linkages, placing most motors near the pelvis to reduce leg inertia and improve movement precision. The 1.84-metre-tall robot supports long-duration reinforcement-learning trials and realistic locomotion studies, helping advance stable, agile humanoid walking for future service and industrial robots.

Robot predicts obstacle movements and plan safer paths

A quantum-inspired optimisation system has been deployed directly on an autonomous mobile robot, enabling faster real-time navigation and decision-making without



Toshiba and MIRISE achieve world's first deployment of a Quantum Inspired Optimization Computer on an Autonomous Mobile Robot (Credit: www.global.toshiba/ww)

relying on cloud computing. It was demonstrated by Toshiba Corporation and MIRISE Technologies during trials in Kawasaki and Nisshin, Japan. The platform integrates

Toshiba's simulated bifurcation machine, an algorithm inspired by quantum computing, into an embedded FPGA to process sensor data and track multiple moving objects simultaneously. Operating at about 23 frames per second, the system improves tracking accuracy and allows robots to predict obstacle movements and plan safer paths. The technology targets applications in factories, smart infrastructure, and autonomous vehicles.

Bioelectronics for energy-efficient AI and next-gen computing memory systems

A bio-hybrid semiconductor memory device by researchers at Pennsylvania State University that combines synthetic DNA with advanced semiconductor materials could significantly improve data storage efficiency in future electronics. The design integrates engineered DNA strands with perovskite semiconductors to create a low-power memristor that simultaneously stores and processes information, similar to neural synapses. The device operates at ultra-low voltages below 0.1V while maintaining stability at temperatures near 250°F, offering improved durability and efficiency. Scientists are exploring bioelectronics to support energy-efficient memory systems for artificial intelligence and next-generation computing.

Soft-robotic glove for swelling relief

A soft-robotic wearable glove developed by researchers at Cornell University electronically compresses the hand to reduce swelling and has been designed to help patients manage oedema at home. The device, called EdemaFlex, uses a net-



EdemaFlex: Soft-robotic glove for swelling relief (Credit: <https://news.cornell.edu>)

work of 37 shape-memory alloy actuators embedded in a textile glove to apply controlled compression from the fingertips toward the wrist, helping move fluid out of swollen tissue. A small printed circuit board controls the compression pattern and intensity through software. where early home-use trials showed reductions in hand swelling, with one participant experiencing up to 25% decrease after a 30-minute session.